

Effects of AI Tutoring on Student Performance

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Abstract

This study examines the impact of AI-based tutoring systems on undergraduate student performance in introductory computer science courses.

Introduction

Artificial intelligence tutoring systems have gained significant attention in recent years as a means to provide personalized learning experiences. Previous research has shown mixed results regarding effectiveness.

Methods

We conducted a randomized controlled trial with 200 undergraduate students. Students were randomly assigned to AI tutoring or traditional tutoring groups. Performance was measured using pre-test and post-test scores.

Results

Students in the AI tutoring group showed a 15% improvement in test scores compared to 8% in the control group ($p < 0.01$).
Effect size was moderate ($d = 0.45$).

Discussion

Our findings suggest that AI tutoring systems can effectively supplement traditional instruction in higher education settings.
However, the effect was most pronounced for students with lower initial scores.

References

Smith, J. (2024). AI in Education: A Comprehensive Review. *Journal of EdTech*.